ECE 6010
Blind Source Separation
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1. Generate 1000 points of a Gaussian(2,5) random variable from a Gaussian(0,1). plot the histogram and overlay the pdf on the histogram. estimate the mean and variance.

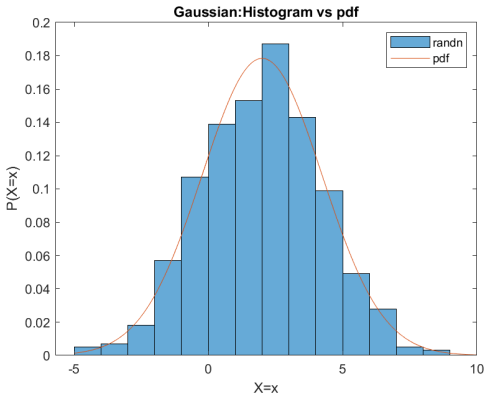


Figure 1: Gaussian with mean 2 and variance 5

2. Generate 1000 points of a uniform(-2,5) random variable from a uniform(0,1). plot the histogram and overlay the pdf on the histogram. estimate the mean and variance.

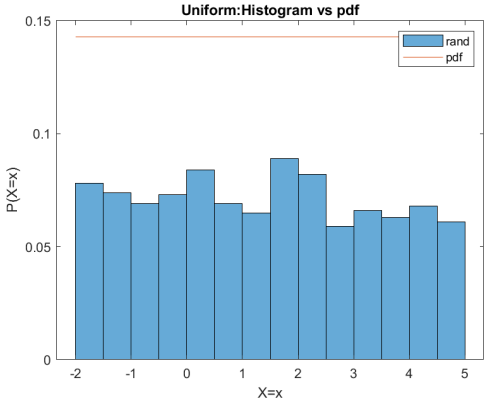


Figure 2: Uniform from -2 to 5

3. write a MATLAB function that will generate N points of Gaussian(mu, SIGMA).

where mu is a vector and SIGMA is a nxn covariance.

estimate the mean vector and covariance matrix using these 1000 generated points. verify that the estimate is resonable of the desired mean and variance.

```
mu_hat1 =
    1.0138
mu_hat2 =
    2.0418
cov_hat =
    0.9284    0.9748
    0.9748    2.8405
```

Figure 3: estimated mean and variance of the function I created

4. plot a scatter plot of the data, plot directions of major and minor axes, plot the contours of the pdf, compare the axes of the ellipses with the eigenvectors of SIGMA, what is the relationship.

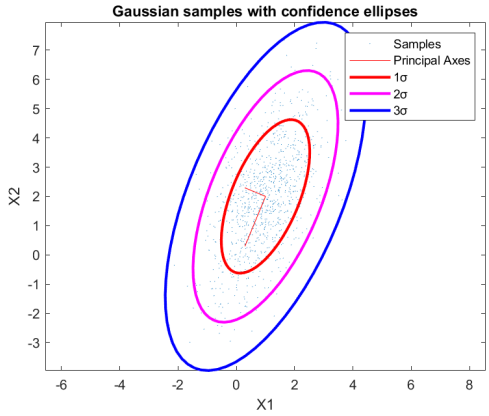


Figure 4: scatter plot of joint gaussian from my functions with contours and major axes

5. Using the Prog1dat. let X(1) be [X1;X2], estimate mu(1) and SIGMA11. let X(2) = [X3;X4], estimate mu(2) and SIGMA22, SIGMA12, SIGMA21.

```
mu_hat2 =
    6.0336    7.0299
cov_22 =
   12.0951    1.5668
    1.5668    8.8043
cov_21 =
    4.7235    2.8393    2.1271    1.0269
    2.8393    9.8718    1.7004    4.0271
    2.1271    1.7004   12.0951    1.5668
    1.0269    4.0271    1.5668    8.8043
```

Figure 5: estimated values of prog data

6. plot the scatter plot of X(1).

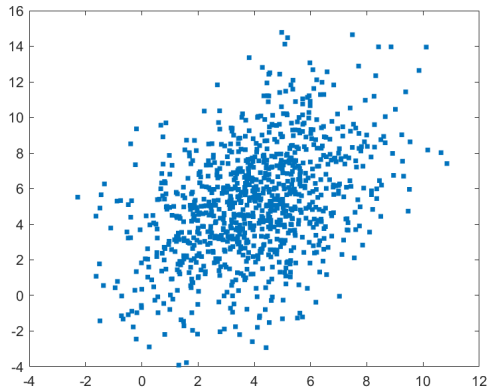


Figure 6: scatter plot of X1 prog data

7. Suppose a measurement of x1, x2, x3 are available. write a function predictx4. what is the estimate of x4, what is the estimate of the variance of the estimate. explicityly state the covariance and mean vectors you use in your computations.

```
SIGMA =
    4.7235    2.8393    2.1271    1.0269
    2.8393    9.8718    1.7004    4.0271
    2.1271    1.7004   12.0951    1.5668
    1.0269    4.0271    1.5668    8.8043
Mu =
    4.0172
    5.2107
    6.0336
    7.0299
x4hat =
    6.3645
sigma4squared =
    7.0796
```

Figure 7: output of the predictx4 function

8. suppose now x1, and x3 are measured and x2, x4 are to be estimated. write a function predict24 which estimates x2, x4, given x1, x3, and which also estimates

the covariance of the estimate. What is the best estimate of x2, and x4. what is the covariance of x2, and x4. what are the estimated values of mu(1), mu(2), SIGMA11, SIGMA21, etc.

```
xhat =
    5.7896
    7.4214
Sigma22 =
    8.1593    3.0231
    3.0231    8.1346
```

Figure 8: output of the predictx24 function

9. write a function called myrandn that will generate a standard Gaussian random variable N(0,1), using uniform U(0,1) random variable and the Box-Muller transformation. Plot a histogram of output data. estimate the mean and variance and compare with desired mean and variance.

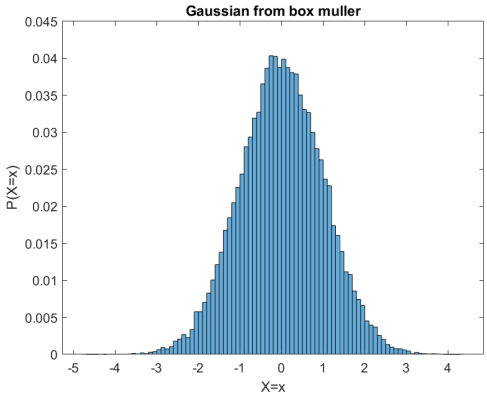


Figure 9: histogram of myrandn with pdf overlay

10. write a function that will generate an exponential(1) random variable, using a transformation of variables. plot a histogram and find the sample mean and compare with theoretical mean.

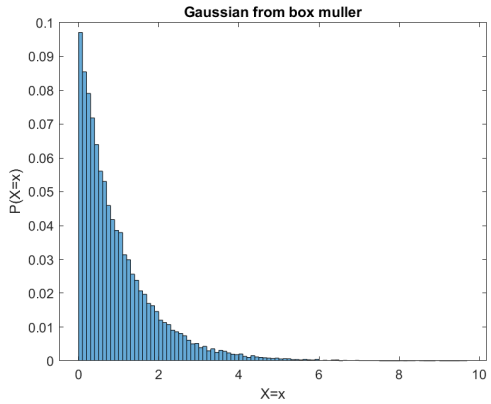


Figure 10: histogram of myexp with pdf overlay

11. X is a uniform(-pi, pi) and let y = sin(X). determine the pdf of Y

    $Y = 1/(2 * pi * \sqrt{(1 - t^2)})$

12. generate 10,000 points of X data and from this, 10,000 points of Y data. plot the histogram with 100 bins. compare the histogram of Y with the pdf of Y.

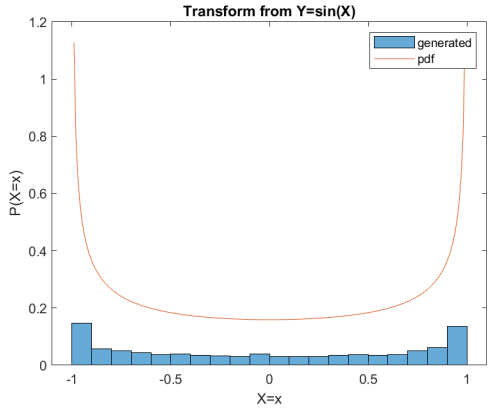


Figure 11: histogram of mysin with pdf overlay

13. make a scatter plot of two independent U(-1,1) random variables.

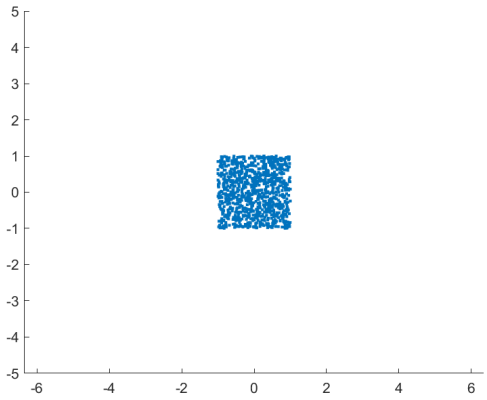


Figure 12: scatter plot of joint Uniform variable

14. transform the data and plot it in the same fashion.

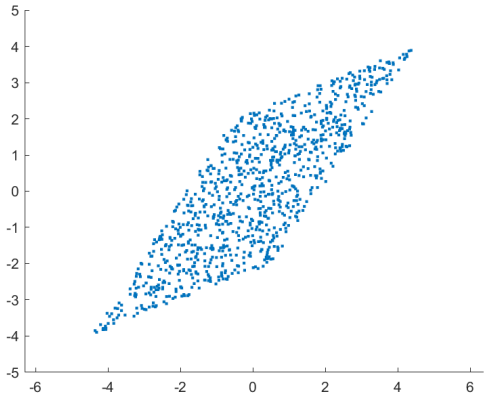


Figure 13: scatter plot of mixed joint uniform variable

15. code the maximum likelihood independent component algorithm. run your uniform transformed data through to see if it corrects. try different values of mu

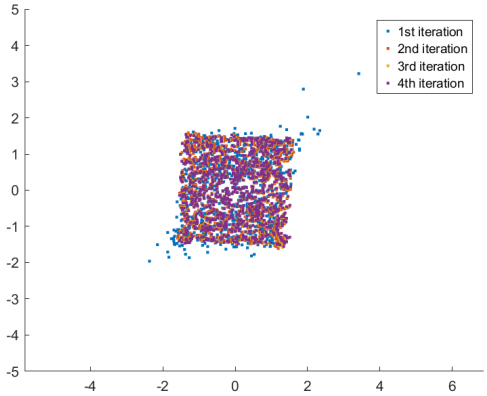


Figure 14: BSS algorithm ran on mixed uniform data

16. use your function to separate the audio from one another in the .mat files. describe how well it works.

    It worked really well! there were some changes in amplitudes while listening but they were distinct and couldn't hear the other audio.

```
function X = gengauss(mu, Sigma, N)
    z = randn(length(mu),N);
    L = chol(Sigma, 'lower');
    X = L*z;
    X = X + mu;
end
```

Listing 1: MATLAB Code

```
clear all; clc; close all; format compact

%% exercise 1
n = 1000;
X = randn(1,n);
X = X*sqrt(5) + 2;
histogram(X, 'Normalization', 'probability');
t = -5:.01:10;
gaus = 1/(sqrt(2*pi*5));
gaus = gaus * exp(-(t - 2).^2/(2*5));
hold on;
plot(t,gaus);
hold off;
legend('randn','pdf');
xlabel('X=x');
ylabel('P(X=x)');
title('Gaussian:Histogram vs pdf');

%% exercise 2
n = 1000;
b = 5;
a = -2;
X = rand(1,n);
X =  X*(b-a) + a;
histogram(X, 'Normalization', 'probability');
t = -2:0.01:5;
uniform = 1/(b-a);
uniform = ones(size(t))*uniform;
hold on;
plot(t,uniform);
hold off;
legend('rand', 'pdf');
xlabel('X=x');
ylabel('P(X=x)');
title('Uniform:Histogram vs pdf')

```

```
%% exercise 3

mu = [1;2];
sigma = [1, 1;1, 3];
N = 1000;
X = gengauss(mu,sigma, N);
mu_hat1 = mean(X(1,:))
mu_hat2 = mean(X(2,:))
cov_hat = cov(X')
% var_hat1 = var(X(1,:))
% var_hat2 = var(X(2,:))
plot(X(1,:), X(2,:), '.','MarkerSize', 1);
axis('equal');
hold on;
[u,v] = eig(inv(sigma));
v1 = u(:,1)/sqrt(v(1,1));
v2 = u(:,2)/sqrt(v(2,2));
plot([mu(1),mu(1)+v1(1)], [mu(2), mu(2) + v1(2)], 'r');
plot([mu(1),mu(1)+v2(1)], [mu(2), mu(2) + v2(2)], 'r');

c_values = [2.30, 6.18, 11.83];  %     1   , 2   , 3
colors = ['r', 'm', 'b'];        % colors for each ellipse

A = inv(sigma);
x0 = mu;

for i = 1:length(c_values)
    c = c_values(i);
    x_ellipse = plotellipse(A, x0, c);
    plot(x_ellipse(1,:), x_ellipse(2,:), colors(i), 'LineWidth', 2);
end

legend('Samples', 'Principal Axes', '', '1   ', '2   ', '3   ');
title('Gaussian samples with confidence ellipses');
xlabel('X1');
ylabel('X2');
hold off;shg;
```

Listing 2: MATLAB Code

```
clear all; close all; clc; format compact;

load prog1dat.mat

X1 = X(1:2,:);
mu_hat11 = mean(X1(1,:));
mu_hat12 = mean(X1(2,:));
mu_hat1 = [mu_hat11, mu_hat12];
cov_11 = cov(X1');
X2 = X(3:4,:);
mu_hat21 = mean(X2(1,:));
```

```
mu_hat22 = mean(X2(2,:));
mu_hat2 = [mu_hat21, mu_hat22]
cov_22 = cov(X2')
cov_21 = cov([X1', X2'])
plot(X1(1,:),X1(2,:),'.','MarkerSize',10);

SIGMA = cov([X(1,:)', X(2,:)', X(3,:)', X(4,:)'])
Mu = mean(X')'
[x4hat, sigma4squared] = predictx4(3, 3.6, 5.2, SIGMA, Mu)
[xhat, Sigma22] = predictx24(5, 7, SIGMA, Mu)
```

Listing 3: MATLAB Code

```
function [x4hat,sigma_4squared] = predictx4(x1, x2, x3, Sigma, Mu)
    X1 = [x1;x2;x3];
    sigma11 = Sigma(1:3,1:3);
    sigma21 = Sigma(4,1:3);
    sigma22 = Sigma(4,4);
    mu1 = Mu(1:3);
    mu2 = Mu(4);
    x4hat = mu2 + sigma21*inv(sigma11)*(X1 - mu1);
    sigma_4squared = sigma22 -sigma21*inv(sigma11)*sigma21';
end
```

Listing 4: MATLAB Code

```
function [xhat,Sigma22] = predictx24(x1, x3, Sigma, Mu)
    X1 = [x1;x3];
    sigma11 = Sigma(1:2:3,1:2:3);
    sigma21 = Sigma(1:2:3,2:2:4);
    sigma22 = Sigma(2:2:4,2:2:4);
    mu1 = Mu(1:2:3);
    mu2 = Mu(2:2:4);
    xhat = mu2 + sigma21*inv(sigma11)*(X1 - mu1);
    Sigma22 = sigma22 -sigma21*inv(sigma11)*sigma21';
end
```

Listing 5: MATLAB Code

```
clear all;clc; close all; format compact

%% gaussian transform
mygaus = [];
for i = 1:30000
    % x = myrandn();
    mygaus = [mygaus, myrandn()];
end

histogram(mygaus, 'Normalization', 'probability');
xlabel('X=x');
ylabel('P(X=x)');
```

```
title('Gaussian from box muller');

%% exponential transform
myexp = [];
for i = 1:30000
    myexp = [myexp, my_exp()];
end

histogram(myexp, 'Normalization', 'probability');
xlabel('X=x');
ylabel('P(X=x)');
title('Gaussian from box muller');

%% sin transform
my_sin = [];
for i = 1:5000
    my_sin = [my_sin, mysin()];
end

histogram(my_sin, 'Normalization', 'probability');
xlabel('X=x');
ylabel('P(X=x)');
title('Transform from Y=sin(X)');
hold on;
t = -1:.01:1;
Y = 1./(2*pi*sqrt(1-t.^2));
plot(t,Y);
hold off;
legend('generated', 'pdf');
shg;
```

Listing 6: MATLAB Code

```
function x = myrandn()
    u1 = rand(1,1);
    u2 = rand(1,1);
    x = sqrt(-2*log(u1))*cos(2*pi*u2);
end
```

Listing 7: MATLAB Code

```
function ex = my_exp()
    u1 = rand(1,1);
    ex = -log(u1);
end
```

Listing 8: MATLAB Code

```
function Y = mysin()
    b = pi;
    a = -pi;
```

```
    X = rand(1,1);
    X =  X*(b-a) + a;
    Y = sin(X);
end
```

Listing 9: MATLAB Code

```
clear all;clc; close all;

N = 1000;
s1 = 2*rand(1,N)-1;
s2 = 2*rand(1,N)-1;
figure();
scatter(s1,s2, '.');
axis([-5 5 -5 5]);
axis("equal");

A = [2.5 2; 1 3];
y = A*[s1;s2];
x1 = y(1,:);
x2 = y(2,:);
figure();
scatter(x1,x2,'.');
axis([-5 5 -5 5]);
axis("equal")

new_y = [];
w0 = eye(2);
w = w0;
N_iterations = 4;
y_iter = y;
figure();
for i = 1:N_iterations
    [y_temp, w] = bss(y, w0, .001, 1);
    w0 = w;
    new_y = y_temp;
    x1 = new_y(1,:);
    x2 = new_y(2,:);
    scatter(x1,x2,'.');
    hold on;
    axis([-5 5 -5 5]);
    axis("equal")
end
legend('1st iteration', '2nd iteration', '3rd iteration', '4th
   iteration');

load x1
load x2
load x3
x = [x1;x2;x3];

```

```
fprintf('separating the data ...\n');
W0 = eye(3);
mu = 0.00025;
Niter = 5;
xiter = x;
for i = 1:Niter
    [y, W] = bss(xiter, W0, mu,2);
    W0 = W;
end
% soundsc(y(3,:));
```

Listing 10: MATLAB Code

```
function [y, B] = bss(x, B0, mu, func)
    I = eye(size(B0));
    y = zeros(size(x));
    xsz = size(x);
    for t = 1:xsz(2)
        y(:,t) = B0 * x(:,t);
        if func == 1
            B = B0 + mu*(I + -y(:,t).^3 * y(:,t)')*B0;
        end
        if func == 2
            B = B0 + mu*(I + -2*tanh(y(:,t)) * y(:,t)') * B0;
        end
        if func == 3
            B = B0 + mu*(I + (tanh(y(:,t)) - y(:,t)) * y(:,t)') * B0
                ;
        end
        B0 = B;
    end
end
```

Listing 11: MATLAB Code